

Ubuntu Linux Administration: Essential Commands

Accessing Linux Locally and Remotely



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Overview



Welcome to the Essentials Course on Ubuntu 20.04

Course Overview

- Accessing the command line
- Using the command line and shortcuts
- Linux file permissions
- Searching for files and content
- Archiving files and compression
- Elevating user privileges

Accessing Linux Locally and Remotely

- Consoles
- SSH and keys
- TightVNC server



Lab Systems



Ubuntu 20.04



The exam can be taken on Ubuntu 20.04 or CentOS 8. We base the complete track on Ubuntu 20.04

- Ubuntu 20.04 LTS - Ubuntu1
- Ubuntu 20.04 LTS - Ubuntu2

We use Vagrant and VirtualBox for these machines, the Vagrantfile is available from Git.

git clone <https://github.com/theurbanpenguin/lfc>



Demo



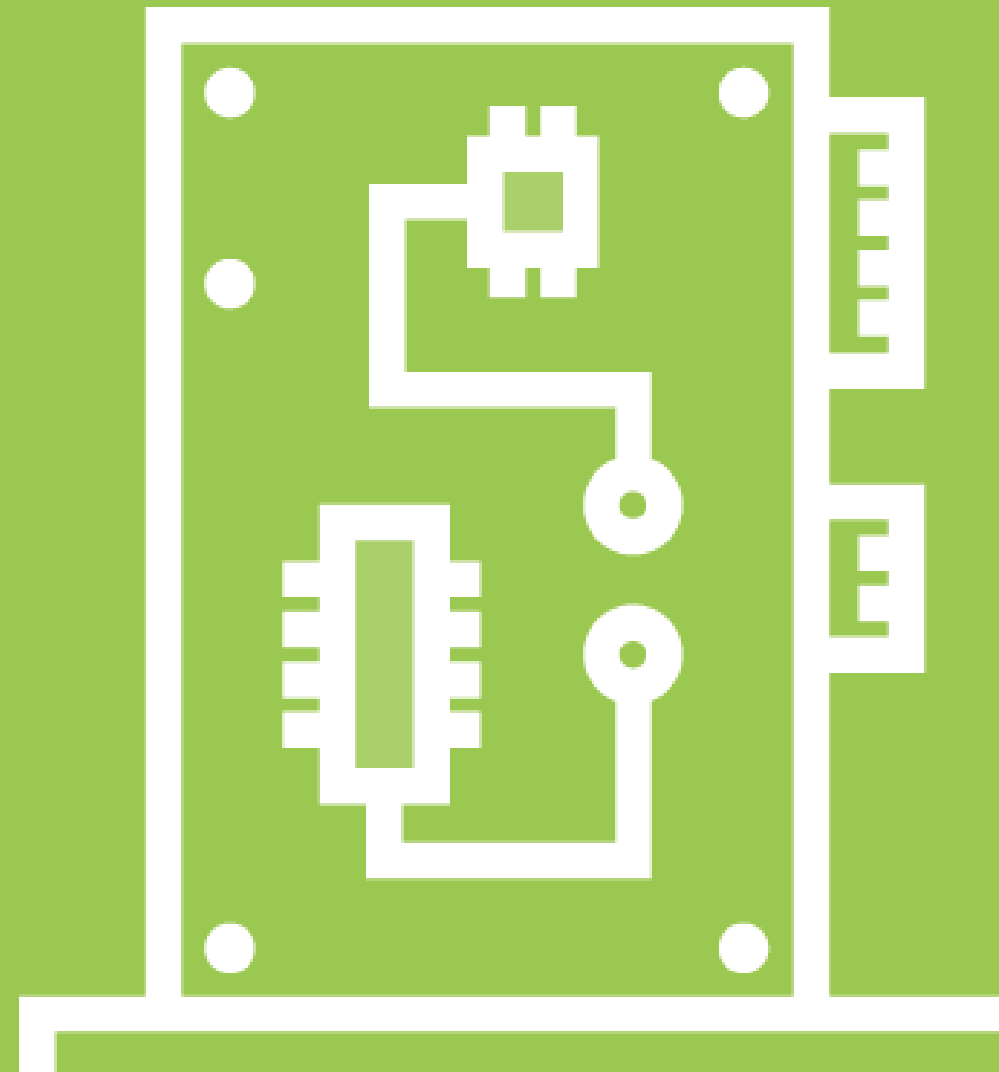
For the first demonstration:

- Start the virtual machines
- Verify connectivity



SSH Access





(SSH) Secure SHell

Command line access to Linux is available via SSH. Windows 10/11 has the OpenSSH client installed by default, as does MacOS and Linux



```
$ ssh vagrant@192.168.56.102
```

```
The authenticity of host '192.168.56.102 (192.168.56.102)' can't be established.  
ECDSA key fingerprint is SHA256:WkogPWXupokxSjJYVHm7CsfKczNozk+EMbBgF3eEFw0.  
Are you sure you want to continue connecting (yes/no/[fingerprint])?
```

```
yes
```

```
vagrant@192.168.56.102's password:
```

Connect to Remote Linux System via SSH

From Ubuntu1 we can connect to Ubuntu2 using SSH. The IP address assigned in the Vagrantfile was 192.168.56.102 for the Ubuntu2 system. If we do not provide a username the current username is used. The password is vagrant, the same as the username.


```
$ ssh-keygen  
$ ssh-copy-id vagrant@192.168.56.102  
$ eval $(ssh-agent)  
$ ssh-add  
$ ssh vagrant@192.168.56.102
```

Key Based Authentication

If we are to connect frequently from Ubuntu1 to Ubuntu2, key-based authentication. may be most convenient. We can generate the key, a passphrase for the key secures it. The public key is copied to the remote user's `authorized_keys` file. To cache the passphrase locally we can run the `ssh-agent` and the add the private key to it.

Demo



Let's look at SSH Access:

- Working on Ubuntu1 we SSH to Ubuntu2



Demo



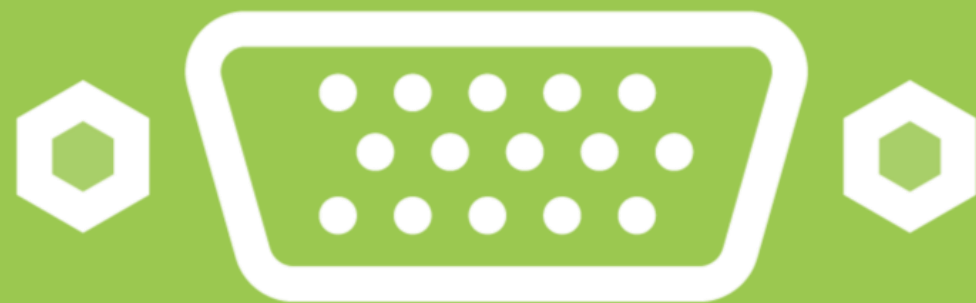
Let's look at SSH Key-Based Authentication:

- Generate key-pair on Ubuntu1
- Copy public key to Ubuntu 2



Remote Graphical Access





VGA

VNC

We can use VNC to connect to remote graphical systems. We are not protected by the encrypted session that we have with SSH, but we can use the GUI programs and desktop.




```
$ sudo apt install -y xfce4 xfce4-goodies
$ sudo apt install -y tightvncserver
$ vncserver
$ vncserver -kill :1
vim ~/.vnc/xstartup
#!/bin/sh
xrdb $HOME/.Xresources
startxfce4&
$ vncserver
```

Working on Ubuntu2

We can install the graphical environment, the vnc server and configure connections

Running **RealVNC** viewer or similar on your host system you can connect to your 192.168.56.102 system on the display number shown, probably :1



Demo



Connecting Graphically:

- Install tightvnc
- Connect from host system



Summary



We have introduced the LFCS Essential Commands course as well as looking at how to connect to the lab systems

- Vagrant
- SSH and consoles
- SSH and key-based authentication
- Remote access with VNC



Up Next:

Using the Linux Command Line Interface

